

Monet-AC Series

Energy Storage Converter

Modbus RTU/TCP communication protocol

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Revised record

Time	Version	Description	Monitor Version	Author
2025-02-25	V0.0.1	Preliminary draft (Based on ModbusRTU protocol V1.0.3, add new communication method Modbus TCP, new IP address setting bit, new port setting bit, new gateway setting bit, new subnet mask setting bit, and new MAC address setting bit.)		Pengxiangyu
2025-02-26	V0.0.2	New Modbus TCP example		Pengxiangyu
2025-04-03	V0.0.3	New 16 function code, new communication failure time		Pengxiangyu
2025-04-18	V0.0.4	Added SN code-related information		Pengxiangyu
2025-05-29	V2.1.0	Improved status information; changed battery float charging voltage and equalization charging voltage to battery charging limit voltage and discharge limit voltage; adjusted primary frequency and voltage regulation parameter ranges; changed charging method to power imbalance control; added parameters for parallel cabinet address, battery SOC, and battery SOC sag; relaxed rated phase voltage and DC-related voltage parameters to adapt to MAC225.	V0.0.4	honghq
2025-08-21	V2.5.0	Add parameters relating to Austrian certification; change 'AC power failure' to 'instantaneous overvoltage or undervoltage in the mains supply' ; widen the range for mains over- and under-frequency; change 'active power change rate' to 'active power change rate on a per-second basis'		honghq
2025-08-23	V2.6.0	New regulatory parameters and configuration parameters relating to Australian certification		honghq
2025-08-26	V2.6.1	Change the maximum visual range limit to 0–1200; add Dutch A and B to the regulations		honghq
2025-09-11	V2.6.2	Regulations: Add Austria		honghq
2025-09-22	V2.6.3	Increase the observation period for the active frequency control of the parameter frequency, the time constant for the PF-P reactive PT1, and the grid voltage correction factor		honghq
2026-01-09	V2.7.0	Added BMS protocol switching and automatic start-up; relaxed the limits for rated phase voltage, DC bus reference and DC bus overvoltage points; removed function code 05 from remote adjustment; added 'None' , 'National Standard' and 'Latvia' to the regulations		honghq

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1. Introductory

Designed to realize the serial communication between the energy storage converter and the host of communication equipment such as EMS, this protocol allows the converter to be connected to the MODBUS RTU/TCP system.

2. Physical interface

2.1. Communication interface

Modbus RTU uses RS485 interface.

Modbus TCP uses the RJ45 interface.

2.2. Information transmission

Modbus RTU is asynchronous with 1 start bit, 8 data bits, 1 stop bit and no parity bit.

Modbus TCP is transmitted via Ethernet using the TCP protocol.

2.3. Data transmission rate

Modbus RTU Baud rate 9600bps.

Modbus TCP Ethernet rate of 10Mb/s.

2.4. Frame interval time requirement

Communication frame interval of 100ms or more is required.

2.5. Communication mode

Adopting master-slave communication mode, the storage converter plays the role of communication slave and waits for the Modbus host to query.

In Modbus TCP, the storage converter is the server and the Modbus host is the client.

2.6. Modbus node number

The physical address of the converter of IP20 is set by dialing code, the setting range is 0~14, corresponding to MODBUS node number 1~15, the factory default dialing code is set to 0, corresponding to MODBUS node number is 1 by default ;

the module address range of the converter of IP65 is 1~15, corresponding to MODBUS node number 1~15, the factory default is 1, and it is set by telecontrol, please refer to 4.5.1 for details of the setting. When communicating to a single IP65 module, use Modbus node number 160, ID number 3182.

2.7. Modbus TCP Related Parameters

The main parameters of Modbus TCP are IP address (ID:3307-3310), port number (ID:3311), subnet mask (ID:3312-3315), gateway (ID:3316-3319), MAC address (ID:3320-3325). Setting the TCP related parameters must be restarted after powering down the energy storage converter to take effect. The MAC addresses must be different within the same LAN.

2.8. CRC calibration

This protocol uses the standard calculation method of Modbus RTU CRC16, the checksum order is (Low-High), low byte first, then high byte.

- The CRC code is calculated by:

1. Preset a 16-bit register to hexadecimal FFFF (i.e., all 1s); call this register the CRC register;

2. Isolate the first 8-bit binary data (i.e., the first byte of the communication message frame) with the lower 8 bits of the 16-bit CRC register, and place the result in the CRC register;

3. Shift the contents of the CRC register one bit to the right (toward the lower bit) to fill the highest bit with a 0, and check the shifted bit after the right shift;

4. If the shifted out bit is 0: repeat step 3 (shift right one bit again); if the shifted out bit is 1: the CRC register is iso-orthogonal to the polynomial A001 (1010 0000 0000 0001);

5. Repeat steps 3 and 4 until you have shifted right 8 times so that the entire 8-bit data has been processed;

6. Repeat steps 2 through 5 for the next byte of the communication frame;

7. After all the bytes of the communication information frame are calculated according to the above steps, the high and low bytes of the 16-bit CRC register are exchanged;

8. The final CRC register content is: CRC code.

3. Function List

3.1. Modbus TCP Message Header

<i>Transaction Processing Identifier</i>	<i>Protocol Identifier</i>	<i>Length</i>	<i>Unit identifier</i>
<i>2 bin</i>	<i>2 bin</i>	<i>2 bin</i>	<i>1 bin</i>

Transaction Processing Identifier: It can be interpreted as the sequence number of the message, plus one after each communication to distinguish between different messages

Protocol Identifier: 00 00 Indicates ModbusTcp protocol

Length: Indicates the length of the next data in bytes

Unit identifier: Modbus node number

3.2. FC3/FC4 Block Read

Modbus RTU Query Frame:

<i>NODE</i>	<i>FUN</i>	<i>ID(start)</i>	<i>NR</i>	<i>CRC</i>
<i>1 bin</i>	<i>1 bin</i>	<i>2 bin</i>	<i>2 bin</i>	<i>2 bin</i>

Modbus TCP Query Frame:

<i>HEADER</i>	<i>FUN</i>	<i>ID(start)</i>	<i>NR</i>
<i>7 bin</i>	<i>1 bin</i>	<i>2 bin</i>	<i>2 bin</i>

HEADER Modbus TCP message header as described in 3.1

NODE MODBUS node number

FUN Function Code, FC3=3 or FC4=4

ID (start) The first address of the queried data area must be the ID defined in the protocol text.

NR Number of IDs of the queried data area

CRC Checksum, L-H method allowed

Modbus RTU Feedback Frame:

<i>NODE</i>	<i>FUN</i>	<i>NR_BYTE</i>	<i>DATA</i>	<i>CRC</i>
<i>1 bin</i>	<i>1 bin</i>	<i>1 bin</i>	<i>n bin</i>	<i>2 bin</i>

Modbus TCP Feedback Frame:

HEADER	FUN	NR_BYTE	DATA
<i>7 bin</i>	<i>1 bin</i>	<i>1 bin</i>	<i>n bin</i>

HEADER Modbus TCP message header as described in 3.1

NODE MODBUS node number

FUN Function Code, FC3=3 or FC4=4

NR_BYTE Number of bytes of feedback data DATA, equal to twice NR

DATA Feedback data area, length equal to NR_BYTE

CRC Checksum, L-H method allowed

3.3. FC5 Single coil write

Modbus RTU Query Frame:

NODE	FUN	ID	DATA	CRC
<i>1 bin</i>	<i>1 bin</i>	<i>2 bin</i>	<i>2bin</i>	<i>2 bin</i>

Modbus TCP Query Frame:

HEADER	FUN	ID	DATA
<i>7 bin</i>	<i>1 bin</i>	<i>2 bin</i>	<i>2 bin</i>

HEADER Modbus TCP message header as described in 3.1

NODE MODBUS node number

FUN Function Code, FC5=5

ID The ID address of the written data must be the ID defined in the protocol text.

DATA Data written, 0xFF00 for ON, 0x0000 for OFF

CRC Checksum, L-H method allowed

Modbus RTU Query Frame:(Return as is)

NODE	FUN	ID	DATA	CRC
<i>1 bin</i>	<i>1 bin</i>	<i>2 bin</i>	<i>2bin</i>	<i>2 bin</i>

Modbus TCP Query Frame:(Return as is)

HEADER	FUN	ID	DATA
<i>7 bin</i>	<i>1 bin</i>	<i>2 bin</i>	<i>2 bin</i>

HEADER Modbus TCP message header as described in 3.1

NODE MODBUS node number

FUN Function Code, FC5=5

ID The ID address of the written data must be the ID defined in the protocol text.

DATA Data written
CRC Checksum, L-H method allowed

3.4. FC6 Word Write

Modbus RTU Query Frame:

NODE	FUN	ID	DATA	CRC
<i>1 bin</i>	<i>1 bin</i>	<i>2 bin</i>	<i>2bin</i>	<i>2 bin</i>

Modbus TCP Query Frame:

HEADER	FUN	ID	DATA
<i>7 bin</i>	<i>1 bin</i>	<i>2 bin</i>	<i>2 bin</i>

HEADER Modbus TCP message header as described in 3.1
 NODE MODBUS node number
 FUN Function Code, FC6=6
 ID The ID address of the written data must be the ID defined in the protocol text.
 DATA Data written
 CRC Checksum, L-H method allowed

Modbus RTU Query Frame:(Return as is)

NODE	FUN	ID	DATA	CRC
<i>1 bin</i>	<i>1 bin</i>	<i>2 bin</i>	<i>2bin</i>	<i>2 bin</i>

Modbus TCP Query Frame:(Return as is)

HEADER	FUN	ID	DATA
<i>7 bin</i>	<i>1 bin</i>	<i>2 bin</i>	<i>2 bin</i>

HEADER Modbus TCP message header as described in 3.1
 NODE MODBUS node number
 FUN Function Code, FC6=6
 ID The ID address of the written data must be the ID defined in the protocol text.
 DATA Data written
 CRC Checksum, L-H method allowed

3.5. FC16 Block Write

Modbus RTU Query Frame:(Return as is)

NODE	FUN	ID(start)	NR	NR_BYTE	DATA	CRC
<i>1 bin</i>	<i>1 bin</i>	<i>2 bin</i>	<i>2 bin</i>	<i>1 bin</i>	<i>n bin</i>	<i>2 bin</i>

Modbus TCP Query Frame:(Return as is)

HEADER	FUN	ID(start)	NR	NR_BYTE	DATA
<i>7 bin</i>	<i>1 bin</i>	<i>2 bin</i>	<i>2 bin</i>	<i>1 bin</i>	<i>n bin</i>

HEADER Modbus TCP message header as described in 3.1

NODE MODBUS node number

FUN Function Code, FC16=16

ID(start) The first address of the ID of the data area to be written must be the ID defined in the protocol text.

NR Number of IDs written to the data area, the legal range is 1 to 123, and there can be no undefined IDs in the coverage area.

NR_BYTE Number of bytes of feedback data DATA, equal to twice NR

DATA Write to data area, length equal to NR_BYTE

CRC Checksum, L-H method allowed

Modbus RTU Feedback Frame:

NODE	FUN	ID(start)	NR	CRC
<i>1 bin</i>	<i>1 bin</i>	<i>2 bin</i>	<i>2 bin</i>	<i>2 bin</i>

Modbus TCP Feedback Frame:

HEADER	FUN	ID(start)	NR
<i>7 bin</i>	<i>1 bin</i>	<i>2 bin</i>	<i>2 bin</i>

HEADER Modbus TCP message header as described in 3.1

NODE MODBUS node number

FUN Function Code, FC16=16

ID(start) The first address of the ID of the data area to be written must be the ID defined in the protocol text.

NR Number of IDs written to the data area, the legal range is 1 to 123, and there can be no undefined IDs in the coverage area.

CRC Checksum, L-H method allowed

3.6. Error return frame

Modbus RTU Return Frame:

NODE	FUN	ERROR	CRC
<i>1 bin</i>	<i>1 bin</i>	<i>1 bin</i>	<i>2 bin</i>

Modbus TCP Return Frame:

HEADER	FUN	ERROR
<i>7 bin</i>	<i>1 bin</i>	<i>1 bin</i>

HEADER	Modbus TCP message header as described in 3.1
NODE	MODBUS node number
FUN	Function number, the highest position of the function number 1, indicates an exception
ERROR	Fault Code Error code 01 = Unknown function code, received function code is not supported. Error code 02 = Address error, the received address (ID) is out of range. Error code 03 = Data error, received data format not recognized, CRC checksum error. Error code 06 = device busy, data not ready
CRC	Checksum, L-H method allowed

4. Protocol text

4.1. Basic Product Information - Function 03 or 04

<i>ID</i>	<i>Byte</i>	<i>Item</i>	<i>Value Type</i>	<i>Format</i>
1000	2	ARM major version number	UINT16	HEX
1001	2	ARM minor version number	UINT16	HEX
1002	2	ARM Version Revision Number	UINT16	HEX
1003	2	ARM version protocol number	UINT16	HEX
1004	2	DSP major version number	UINT16	HEX
1005	2	DSP minor version number	UINT16	HEX
1006	2	DSP Version Revision Number	UINT16	HEX
1007	2	CPLD software version number	UINT16	HEX
1009	2	Country Name 1 Country Name 2	UINT16	ASCII
1010	2	Country Name 3 Country Name 4	UINT16	ASCII
1011	2	Country Name 5 Country Name 6	UINT16	ASCII
1012	2	Country Name 7 Country Name 8	UINT16	ASCII
1013	2	Country Name 9 Country Name 10	UINT16	ASCII
1014	2	Country Name 11 Country Name 12	UINT16	ASCII
1015	2	Country Name 13 Country Name 14	UINT16	ASCII
1016	2	Country Name 15 Country Name 16	UINT16	ASCII
1017	2	Country Name 17 Country Name 18	UINT16	ASCII
1018	2	Country Name 19 Country Name 20	UINT16	ASCII
1019	2	Country Name 21 Country Name 22	UINT16	ASCII
1020	2	Country Name 23 Country Name 24	UINT16	ASCII

1021	2	Model 1 Model 2	UINT16	ASCII
1022	2	Model 3 Model 4	UINT16	ASCII
1023	2	Model 5 Model 6	UINT16	ASCII
1024	2	Model 7 Model 8	UINT16	ASCII
1025	2	Model 9 Model 10	UINT16	ASCII
1026	2	Model 11 Model 12	UINT16	ASCII
1027	2	Model 13 Model 14	UINT16	ASCII
1028	2	Model 15 Model 16	UINT16	ASCII
1029	2	Model 17 Model 18	UINT16	ASCII
1030	2	Model 19 Model 20	UINT16	ASCII
1031	2	Model 21 Model 22	UINT16	ASCII
1032	2	Model 23 Model 24	UINT16	ASCII
1033	2	Serial No. 1 Serial No. 2	UINT16	ASCII
1034	2	Serial No. 3 Serial No. 4	UINT16	ASCII
1035	2	Serial No. 5 Serial No. 6	UINT16	ASCII
1036	2	Serial No. 7 Serial No. 8	UINT16	ASCII
1037	2	Serial No. 9 Serial No. 10	UINT16	ASCII
1038	2	Serial No. 11 Serial No. 12	UINT16	ASCII
1039	2	Serial No. 13 Serial No. 14	UINT16	ASCII
1040	2	Serial No. 15 Serial No. 16	UINT16	ASCII
1041	2	Serial No. 17 Serial No. 18	UINT16	ASCII
1042	2	Serial No. 19 Serial No. 20	UINT16	ASCII

1043	2	Serial No. 21 Serial No. 22	UINT16	ASCII
1044	2	Serial No. 23 Serial No. 24	UINT16	ASCII

4.2. Analog Information (Telemetry) - Function 03 or 04

4.2.1. System analog

ID	Byte	Item	Value	Value Type	Format
1050	2	Grid voltage A(AB)	2200corresponds to220.0V	UINT16	HEX
1051	2	Grid voltage B(BC)	2200corresponds to220.0V	UINT16	HEX
1052	2	Grid voltage C(CA)	2200corresponds to220.0V	UINT16	HEX
1053	2	Grid current A	1000corresponds to100.0A	UINT16	HEX
1054	2	Grid current B	1000corresponds to100.0A	UINT16	HEX
1055	2	Grid current C	1000corresponds to100.0A	UINT16	HEX
1056	2	Grid frequency A	5000corresponds to50.00Hz	UINT16	HEX
1057	2	Grid frequency B	5000corresponds to50.00Hz	UINT16	HEX
1058	2	Grid frequency C	5000corresponds to50.00Hz	UINT16	HEX
1059	2	Total power factor	100corresponds to1.00	INT16	HEX
1060	2	Total active power	1000corresponds to100.0kW	INT16	HEX
1061	2	Total reactive power	1000corresponds to100.0kvar	INT16	HEX
1062	2	Total apparent power	1000corresponds to100.0kVA	UINT16	HEX
1063	2	DC voltage	1000corresponds to100.0V	UINT16	HEX
1064	2	DC current	1000corresponds to100.0A	INT16	HEX
1065	2	DC power	1000corresponds to100.0kW	INT16	HEX
1066	2	Busbar voltage	1000corresponds to100.0V	UINT16	HEX
1067	2	Positive bus voltage	1000corresponds to100.0V	UINT16	HEX
1068	2	Negative bus voltage	1000corresponds to100.0V	UINT16	HEX
1069	2	Balanced bridge current	1000corresponds to100.0A	INT16	HEX
1070	2	Ground-to-negative voltage	1000corresponds to100.0V	INT16	HEX
1071	2	Positive to ground impedance	100corresponds to100kΩ	UINT16	HEX

1072	2	Negative Impedance to Ground	100corresponds to100k Ω	UINT16	HEX
1073	2	Leakage Current	100corresponds to1.00A	UINT16	HEX
1074	2	Power Tube Temperature	100corresponds to100°C	INT16	HEX
1075	2	Balance Bridge Temperature	100corresponds to100°C	INT16	HEX
1076	2	Ambient temperature	100corresponds to100°C	INT16	HEX
1077	2	Converter efficiency	1000corresponds to100.0%	UINT16	HEX
1078	2	Power Factor A	100corresponds to1.00	INT16	HEX
1079	2	Power Factor B	100corresponds to1.00	INT16	HEX
1080	2	Power factor C	100corresponds to1.00	INT16	HEX
1081	2	Active power A	1000corresponds to100.0kW	INT16	HEX
1082	2	Active power B	1000corresponds to100.0kW	INT16	HEX
1083	2	Active power C	1000corresponds to100.0kW	INT16	HEX
1084	2	Reactive power A	1000corresponds to100.0 kvar	INT16	HEX
1085	2	Reactive power B	1000corresponds to100.0 kvar	INT16	HEX
1086	2	Reactive power C	1000corresponds to100.0 kvar	INT16	HEX
1087	2	Apparent power A	1000corresponds to100.0kVA	UINT16	HEX
1088	2	Apparent power B	1000corresponds to100.0kVA	UINT16	HEX
1089	2	Apparent power C	1000corresponds to100.0kVA	UINT16	HEX
1090	2	Inverter voltage A(AB)	2200corresponds to220.0V	UINT16	HEX
1091	2	Inverter voltage B(BC)	2200corresponds to220.0V	UINT16	HEX
1092	2	Inverter voltage C(CA)	2200corresponds to220.0V	UINT16	HEX
1093	2	Inverter current A	1000corresponds to100.0A	UINT16	HEX
1094	2	Inverter current B	1000corresponds to100.0A	UINT16	HEX
1095	2	Inverter current C	1000corresponds to100.0A	UINT16	HEX
1096	2	Reserved	2200corresponds to220.0V	INT16	HEX
1097	2	Reserved	2200corresponds to220.0V	INT16	HEX
1098	2	Reserved	2200corresponds to220.0V	INT16	HEX
1099	2	Reserved	1000corresponds to100.0A	INT16	HEX

1100	2	Reserved	1000corresponds to100.0A	INT16	HEX
1101	2	Reserved	1000corresponds to100.0A	INT16	HEX
1102	2	Reserved	1corresponds to1	UINT16	HEX
1103	2	Reserved	1corresponds to1	UINT16	HEX
1104	2	Reserved	1corresponds to1	UINT16	HEX
1105 1106	4	Total charge ⁽¹⁾	1000corresponds to100.0kWh	UINT32	HEX Hi-Lo
1107 1108	4	Total discharge ⁽¹⁾	1000corresponds to100.0kWh	UINT32	HEX Hi-Lo

Notes:

(1) The total charging and discharging amount accumulated is the total power value from this power-up to the current moment, and the initial value is reset to 1 for the next power-up.

4.2.2. Telemetry Example

Example: Modbus node number 01, read telemetry data: 3 analogue values starting from ID 1050, data order H-L, checksum order L-H.

Modbus RTU:

Telemetry (HEX): 01 03 04 1A 00 03 25 3C

Where: 01 - Modbus number 01; 03 - function code 03;
04 1A - starting ID of 1050; 00 03 - quantity 3;
25 3C - CRC16 checksum

Answer (HEX): 01 03 06 08 98 08 98 08 84 04

Where: 01 - Modbus number 01; 03 - function code 03;
06 - Number of data bytes; 08 98-220V Grid voltage A (AB)
08 98-220V Grid voltage B (BC) 08 98-220V Grid voltage C (CA)
84 04-CRC16 checksum

Modbus TCP:

Telemetry (HEX): 00 06 00 00 00 06 01 03 04 1A 00 03

Where: 00 06 - transaction identification; 00 00 - protocol identifier;
00 06 - length;
01 - Unit identifier (Modbus node number) 01;
03 - Function Code 03 04 1A- Starting ID is 1050
00 03 - Quantity 3

Answer (HEX): 00 06 00 00 00 09 01 03 06 08 98 08 98 08 98 08 98

Where:

- 00 06 - transaction identification; 00 00 - protocol identifier;
- 00 09 - length;
- 01 - Unit identifier (Modbus node number) 01
- 03 - Function Code 03 06 - Number of data bytes
- 08 98-220V Grid voltage A(AB) 08 98-220V Grid voltage B (BC)
- 08 98-220V Grid voltage C (CA)

4.3. Alarm volume (telemetry) - Function 03 or 04

4.3.1. Alarm message 1

<i>ID</i>	<i>Bit</i>	<i>Item</i>	<i>Value</i>
2050	<i>Bit 0</i>	<i>Positive bus over-pressure</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 1</i>	<i>Negative bus over-voltage</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 2</i>	<i>DC bus over-voltage</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 3</i>	<i>Bus half-voltage unbalance</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 4</i>	<i>DC bus short circuit</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 5</i>	<i>DC over-current</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 6</i>	<i>Balance bridge over-current</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 7</i>	<i>DC Voltage Reverse</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 8</i>	<i>Low DC Voltage</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 9</i>	<i>DC voltage high</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit10</i>	<i>Abnormal insulation impedance</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 11</i>	<i>PV power low shutdown</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 12</i>		
	<i>Bit 13</i>		
	<i>Bit 14</i>		
	<i>Bit 15</i>		

4.3.2. Alarm message 2

<i>ID</i>	<i>Bit</i>	<i>Item</i>	<i>Value</i>
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2051	Bit 0	Abnormal phase-A inverter voltage	"0: Normal" "1:Malfunction"
	Bit 1	Abnormal phase-B inverter voltage	"0: Normal" "1:Malfunction"
	Bit 2	Abnormal phase-C inverter voltage	"0: Normal" "1:Malfunction"
	Bit 3	A-Phase inverter voltage direct current abnormal	"0: Normal" "1:Malfunction"
	Bit 4	B-phase inverter voltage direct current abnormal	"0: Normal" "1:Malfunction"
	Bit 5	C-Phase inverter voltage direct current abnormal	"0: Normal" "1:Malfunction"
	Bit 6	A-Phase output overload shutdown	"0: Normal" "1:Malfunction"
	Bit 7	B-phase output overload shutdown	"0: Normal" "1:Malfunction"
	Bit 8	C-phase output overload shutdown	"0: Normal" "1:Malfunction"
	Bit 9	A-phase output over-current	"0: Normal" "1:Malfunction"
	Bit10	B-phase output over-current	"0: Normal" "1:Malfunction"
	Bit 11	C-phase output over-current	"0: Normal" "1:Malfunction"
	Bit 12	A-phase output short circuit	"0: Normal" "1:Malfunction"
	Bit 13	B-phase output short circuit	"0: Normal" "1:Malfunction"
	Bit 14	C-phase output short circuit	"0: Normal" "1:Malfunction"
	Bit 15	Inverter phase out of synchronization	"0: Normal" "1:Malfunction"

4.3.3. Alarm message 3

<i>ID</i>	<i>Bit</i>	<i>Item</i>	<i>Value</i>
2052	<i>Bit 0</i>	<i>Grid over-voltage</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 1</i>	<i>Grid under-voltage</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 2</i>	<i>Grid over-frequency</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 3</i>	<i>Grid under-frequency</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 4</i>	<i>Islanding protection</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 5</i>	<i>Grid wrong phase</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 6</i>	<i>AC Power Failure</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 7</i>	<i>AC current limiting shutdown</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 8</i>	<i>Parallel Cable Fault</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 9</i>	<i>Carrier Synchronization Fault</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit10</i>	<i>Inverter Synchronization Fault</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 11</i>	<i>Parallel Communication Fault</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 12</i>	<i>AC Fuse Failure</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 13</i>	<i>Power Tube Over Temperature</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 14</i>	<i>Power Supply Fault</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 15</i>	<i>Leakage current fault</i>	<i>"0: Normal" "1:Malfunction"</i>

4.3.4. Alarm message 4

<i>ID</i>	<i>Bit</i>	<i>Item</i>	<i>Value</i>
2053	<i>Bit 0</i>	<i>DC Precharge Fault</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 1</i>	<i>AC Precharge Fault</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 2</i>	<i>DC Relay Short Circuit</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 3</i>	<i>DC Relay Open Circuit</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 4</i>	<i>AC Phase A-phase Relay Short Circuit</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 5</i>	<i>AC B-phase relay short circuit</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 6</i>	<i>AC C-phase relay short circuit</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 7</i>	<i>AC A-phase relay open</i>	<i>"0: Normal" "1:Malfunction"</i>

	<i>Bit 8</i>	<i>AC B-phase relay open</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 9</i>	<i>AC C-phase relay open</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit10</i>	<i>A-phase bridge arm straight through</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 11</i>	<i>B-phase bridge arm through</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 12</i>	<i>C-phase bridge arm through</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 13</i>		
	<i>Bit 14</i>		
	<i>Bit 15</i>		

4.3.5. Alarm message 5

<i>ID</i>	<i>Bit</i>	<i>Item</i>	<i>Value</i>
2054	<i>Bit 0</i>	<i>Grid Current Zero Bias Abnormal</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 1</i>	<i>Inverter current zero bias anomaly</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 2</i>	<i>Inverter current direct current zero bias anomaly</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 3</i>	<i>Zero bias DC current anomaly</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 4</i>	<i>Balance bridge current zero bias anomaly</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 5</i>	<i>Leakage current zero bias anomaly</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 6</i>	<i>2.5V reference abnormal</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 7</i>		
	<i>Bit 8</i>		
	<i>Bit 9</i>		
	<i>Bit10</i>		
	<i>Bit 11</i>		
	<i>Bit 12</i>		
	<i>Bit 13</i>		
	<i>Bit 14</i>		
	<i>Bit 15</i>		

4.3.6. Alarm message 6

<i>ID</i>	<i>Bit</i>	<i>Item</i>	<i>Value</i>
2055	<i>Bit 0</i>	<i>A-phase output overload alarm</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 1</i>	<i>B-phase output overload alarm</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 2</i>	<i>C-Phase output overload alarm</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 3</i>	<i>Low voltage ride-through</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 4</i>	<i>High voltage ride-through</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 5</i>	<i>Balanced bridge wave-by-wave current limit alarm</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 6</i>	<i>Balance bridge over temperature</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 7</i>	<i>High ambient temperature</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 8</i>	<i>Over Temperature Derating</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 9</i>	<i>DC lightning protection fault</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit10</i>	<i>AC lightning protection fault</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 11</i>	<i>Fan fault 1</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 12</i>	<i>Fan fault 2</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 13</i>	<i>Battery over-charge</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 14</i>	<i>Battery over-discharge</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 15</i>		

4.3.7. Alarm message 7

<i>ID</i>	<i>Bit</i>	<i>Item</i>	<i>Value</i>
2056	<i>Bit 0</i>	<i>Internal Communication Failure</i>	<i>"0: No" "1: Yes"</i>
	<i>Bit 1</i>	<i>EMS communication failure</i>	<i>"0: No" "1: Yes"</i>
	<i>Bit 2</i>	<i>BMS communication failure</i>	<i>"0: No" "1: Yes"</i>
	<i>Bit 3</i>	<i>BMS fault</i>	<i>"0: No" "1: Yes"</i>
	<i>Bit 4</i>	<i>Dry contact 1 fault</i>	<i>"0: No" "1: Yes"</i>
	<i>Bit 5</i>	<i>Dry contact 2 fault</i>	<i>"0: No" "1: Yes"</i>
	<i>Bit 6</i>		

	<i>Bit 7</i>		
	<i>Bit 8</i>		
	<i>Bit 9</i>		
	<i>Bit10</i>		
	<i>Bit 11</i>		
	<i>Bit 12</i>		
	<i>Bit 13</i>		
	<i>Bit 14</i>		
	<i>Bit 15</i>		

4.4. Status Quantity (Telecommand) - Function 03 or 04

4.4.1. Status message 1

<i>ID</i>	<i>Bit</i>	<i>Item</i>	<i>Value</i>
2057	<i>Bit 0</i>	<i>DC Pre-Charge</i>	<i>"0: Disconnected" "1: Connected"</i>
	<i>Bit 1</i>	<i>AC Soft Start</i>	<i>"0: Disconnected" "1: Connected"</i>
	<i>Bit 2</i>	<i>DC Relay</i>	<i>"0: Disconnected" "1: Connected"</i>
	<i>Bit 3</i>	<i>AC Relay</i>	<i>"0: Disconnected" "1: Connected"</i>
	<i>Bit 4</i>	<i>Off-grid</i>	<i>"0: On-Grid" "1: Off-Grid"</i>
	<i>Bit 5</i>	<i>Charge and Discharge</i>	<i>"0: Discharging" "1: Charging"</i>
	<i>Bit 6</i>	<i>Operation Status</i>	<i>"0: Stop" "1: Run"</i>
	<i>Bit 7</i>	<i>Fault status</i>	<i>"0: No" "1: Yes"</i>
	<i>Bit 8</i>	<i>Master-slave machine</i>	<i>"0: Slave machine" "1: Master machine"</i>
	<i>Bit 9</i>	<i>Master-slave cabinet</i>	<i>"0: Slave cabinet" "1: Master cabinet"</i>
	<i>Bit10</i>	<i>Standby</i>	<i>"0: No" "1: Standby"</i>
	<i>Bit 11</i>	<i>Shutdown</i>	<i>"0: No" "1: Shutdown"</i>
	<i>Bit 12</i>	<i>EPO</i>	<i>"0: No" "1: Emergency stop"</i>
	<i>Bit 13</i>		
	<i>Bit 14</i>		
	<i>Bit 15</i>	<i>Parameter initialization</i>	<i>"0: Not completed" "1: Completed"</i>

4.4.2. Example of telecommunication

Example: Modbus node 01, reading telecommunication message, start ID 2050, quantity 1, fault is insulation impedance abnormality, data sequence H-L, checksum sequence L-H.

Modbus RTU:

Telecommand (HEX): 01 03 08 02 00 01 27 AA

Answer (HEX): 01 03 02 04 00 BA 84

Modbus TCP:

Telecommand (HEX): 00 03 00 00 00 06 01 03 08 02 00 01

Answer (HEX): 00 03 00 00 00 05 01 03 02 04 00

4.5. Parameter setting (remote control) - Function 03 or 04 or 06 or 16

4.5.1. Control parameter setting

<i>ID</i>	<i>Byte</i>	<i>Item</i>	<i>Value</i>	<i>Value Type</i>	<i>Format</i>
3050	2	Total active power	-1200~1200 (-120%~120%) (default 50)	INT16	HEX
3051	2	Total reactive power	-1200~1200 (-120%~120%) (default 0)	INT16	HEX
3052	2	Total power factor	-1000~1000 (-1~1) (default 1000)	INT16	HEX
3053	2	A-phase active power	-1200~1200 (-120%~120%) (default 50)	INT16	HEX
3054	2	B-phase active power	-1200~1200 (-120%~120%) (default 50)	INT16	HEX
3055	2	Phase C active power	-1200~1200 (-120%~120%) (default 50)	INT16	HEX
3056	2	A-phase reactive power	-1200~1200 (-120%~120%) (default 0)	INT16	HEX
3057	2	B-phase reactive power	-1200~1200 (-120%~120%) (default 0)	INT16	HEX
3058	2	Phase C reactive power	-1200~1200 (-120%~120%) (default 0)	INT16	HEX

3059	2	A-phase power factor	-1000~1000 (-1~1) (default 1000)	INT16	HEX
3060	2	B-phase power factor	-1000~1000 (-1~1) (default 1000)	INT16	HEX
3061	2	Phase C power factor	-1000~1000 (-1~1) (default 1000)	INT16	HEX
3062	2	Active rate of change	1~30000 (1%/s~30000%/s) (default 2000)	UINT16	HEX
3063	2	Grid over-voltage level 1	105~150 (105%~150%) (default 110)	UINT16	HEX
3064	2	Grid over-voltage level 2	105~150 (105%~150%) (default 120)	UINT16	HEX
3065	2	Grid over-voltage class 3	105~150 (105%~150%) (default 130)	UINT16	HEX
3066	2	Grid over-voltage level 4	105~150 (105%~150%) (default 135)	UINT16	HEX
3067	2	Grid over-voltage level 5	105~150 (105%~150%) (default 140)	UINT16	HEX
3068	2	Grid over-voltage recovery	105~150 (105%~150%) (default 108)	UINT16	HEX
3069	2	Grid under-voltage level 1	10~95 (10%~95%) (default 85)	UINT16	HEX
3070	2	Grid under-voltage class 2	10~95 (10%~95%) (default 80)	UINT16	HEX
3071	2	Grid under-voltage 3	10~95 (10%~95%) (default 60)	UINT16	HEX
3072	2	Grid under-voltage level 4	10~95 (10%~95%) (default 40)	UINT16	HEX
3073	2	Grid under-voltage level 5	10~95 (10%~95%) (default 20)	UINT16	HEX
3074	2	Grid under-voltage recovery	10~95 (10%~95%) (default 87)	UINT16	HEX
3075	2	Grid over-frequency level 1	1~500 (+0.01 ~ +5Hz) (default 100)	UINT16	HEX

3076	2	Grid over-frequency level 2	1~500 (+0.01 ~ +5Hz) (default 200)	UINT16	HEX
3077	2	Grid over-frequency 3	1~500 (+0.01 ~ +5Hz) (default 300)	UINT16	HEX
3078	2	Grid over-frequency level 4	1~500 (+0.01 ~ +5Hz) (default 400)	UINT16	HEX
3079	2	Grid over-frequency level 5	1~500 (+0.01 ~ +5Hz) (default 500)	UINT16	HEX
3080	2	Grid over-frequency recovery	1~500 (+0.01 ~ +5Hz) (default 80)	UINT16	HEX
3081	2	Grid under-frequency level 1	-500~1 (-5~ -0.01Hz) (default -100)	INT16	HEX
3082	2	Grid under-frequency 2	-500~1 (-5~ -0.01Hz) (default -200)	INT16	HEX
3083	2	Grid under-frequency 3	-500~1 (-5~ -0.01Hz) (default -300)	INT16	HEX
3084	2	Grid under-frequency level 4	-500~1 (-5~ -0.01Hz) (default -400)	INT16	HEX
3085	2	Grid under-frequency level 5	-500~1 (-5~ -0.01Hz) (default -500)	INT16	HEX
3086	2	Grid under-frequency recovery	-500~1 (-5~ -0.01Hz) (default -80)	INT16	HEX
3087 3088	4	Grid over-voltage protection time	0~3600000 (ms) (default 15000)	UINT32	HEX Hi-Lo
3089 3090	4	Grid over-voltage secondary protection time	0~3600000 (ms) (default 2000)	UINT32	HEX Hi-Lo
3091 3092	4	Grid over-voltage three-level protection time	0~3600000 (ms) (default 100)	UINT32	HEX Hi-Lo
3093 3094	4	Grid over-voltage four-level protection time	0~3600000 (ms) (default 100)	UINT32	HEX Hi-Lo
3095	4	Grid over-voltage	0~3600000 (ms)	UINT32	HEX

3096		<i>five-level protection time</i>	<i>(default 20)</i>		<i>Hi-Lo</i>
3097 3098	4	<i>Grid under-voltage protection time</i>	<i>0~3600000 (ms)</i> <i>(default 5000)</i>	<i>UINT32</i>	<i>HEX</i> <i>Hi-Lo</i>
3099 3100	4	<i>Grid under-voltage secondary protection time</i>	<i>0~3600000 (ms)</i> <i>(default 3000)</i>	<i>UINT32</i>	<i>HEX</i> <i>Hi-Lo</i>
3101 3102	4	<i>Grid under-voltage three-level protection time</i>	<i>0~3600000 (ms)</i> <i>(default 2000)</i>	<i>UINT32</i>	<i>HEX</i> <i>Hi-Lo</i>
3103 3104	4	<i>Grid under-voltage four-level protection time</i>	<i>0~3600000 (ms)</i> <i>(default 1500)</i>	<i>UINT32</i>	<i>HEX</i> <i>Hi-Lo</i>
3105 3106	4	<i>Grid under-voltage five-level protection time</i>	<i>0~3600000 (ms)</i> <i>(default 1000)</i>	<i>UINT32</i>	<i>HEX</i> <i>Hi-Lo</i>
3107 3108	4	<i>Grid over-frequency first level protection time</i>	<i>0~3600000 (ms)</i> <i>(default 30000)</i>	<i>UINT32</i>	<i>HEX</i> <i>Hi-Lo</i>
3109 3110	4	<i>Grid over-frequency secondary protection time</i>	<i>0~3600000 (ms)</i> <i>(default 10000)</i>	<i>UINT32</i>	<i>HEX</i> <i>Hi-Lo</i>
3111 3112	4	<i>Grid over-frequency three-level protection time</i>	<i>0~3600000 (ms)</i> <i>(default 5000)</i>	<i>UINT32</i>	<i>HEX</i> <i>Hi-Lo</i>
3113 3114	4	<i>Grid over-frequency four-level protection time</i>	<i>0~3600000 (ms)</i> <i>(default 1000)</i>	<i>UINT32</i>	<i>HEX</i> <i>Hi-Lo</i>
3115 3116	4	<i>Grid over-frequency five-level protection time</i>	<i>0~3600000 (ms)</i> <i>(default 100)</i>	<i>UINT32</i>	<i>HEX</i> <i>Hi-Lo</i>
3117 3118	4	<i>Grid under-frequency first level protection time</i>	<i>0~3600000 (ms)</i> <i>(default 30000)</i>	<i>UINT32</i>	<i>HEX</i> <i>Hi-Lo</i>

3119 3120	4	Grid under-frequency secondary protection time	0~3600000 (ms) (default 10000)	UINT32	HEX Hi-Lo
3121 3122	4	Grid under-frequency three-level protection time	0~3600000 (ms) (default 5000)	UINT32	HEX Hi-Lo
3123 3124	4	Grid under-frequency four-level protection time	0~3600000 (ms) (default 1000)	UINT32	HEX Hi-Lo
3125 3126	4	Grid under-frequency fifth level protection time	0~3600000 (ms) (default 100)	UINT32	HEX Hi-Lo
3127 3128	4	Grid restoration confirmation time	0~3600000 (ms) (default 30000)	UINT32	HEX Hi-Lo
3129	2	Low penetration reactive current coefficient	0~600(0~6) (default 160)	UINT16	HEX
3130	2	Low penetration active current coefficient	0~100(0~1) (default 100)	UINT16	HEX
3131	2	Low penetration active recovery rate	1~30000(1~30000%/s) (default 100)	UINT16	HEX
3132	2	High penetration reactive current coefficient	0~600(0~6) (default 160)	UINT16	HEX
3133	2	High Throughput Active Coefficient	0~100(0~1) (default 100)	UINT16	HEX
3134	2	Battery Type	0: lithium, 1: lead acid, 2: PV, 3: DC bus (default 0)	UINT16	HEX
3135	2	Battery under-voltage protection point	2000~18000(200~1800V) (default 6000)	UINT16	HEX
3136	2	Battery under-voltage	2000~18000(200~1800V) (default 6200)	UINT16	HEX

		<i>recovery point</i>			
3137	2	<i>Battery over-voltage protection point</i>	<i>2000~18000(200~1800V) (default 9000)</i>	<i>UINT16</i>	<i>HEX</i>
3138	2	<i>Battery over-voltage Recovery Point</i>	<i>2000~18000(200~1800V) (default 8800)</i>	<i>UINT16</i>	<i>HEX</i>
3139	2	<i>Battery charge current limit</i>	<i>0~4000(0~400A) (default 2000)</i>	<i>UINT16</i>	<i>HEX</i>
3140	2	<i>Battery discharge current limit</i>	<i>0~4000(0~400A) (default 2000)</i>	<i>UINT16</i>	<i>HEX</i>
3141	2	<i>Battery float voltage</i>	<i>2000~18000(200~1800V) (default 8700)</i>	<i>UINT16</i>	<i>HEX</i>
3142	2	<i>Battery equalization voltage</i>	<i>2000~18000(200~1800V) (default 8800)</i>	<i>UINT16</i>	<i>HEX</i>
3143	2	<i>DC bus voltage reference</i>	<i>3000~18000(300~1800V) (default 7500)</i>	<i>UINT16</i>	<i>HEX</i>
3144	2	<i>Insulation protection threshold</i>	<i>1~1000(kΩ) (default 100)</i>	<i>UINT16</i>	<i>HEX</i>
3145	2	<i>DC bus over-voltage point</i>	<i>3000~18000(300~1800V) (default 9500)</i>	<i>UINT16</i>	<i>HEX</i>
3146	2	<i>Rated power</i>	<i>100~3000 (10~300kW) (default 500)</i>	<i>UINT16</i>	<i>HEX</i>
3147	2	<i>Rated phase voltage</i>	<i>10~600 (V) (default 230)</i>	<i>UINT16</i>	<i>HEX</i>
3148	2	<i>Rated frequency</i>	<i>50~60 (Hz) (default 50)</i>	<i>UINT16</i>	<i>HEX</i>
3149	2	<i>Grid-side wiring</i>	<i>0: 3-wire, 1: 4-wire (default 0)</i>	<i>UINT16</i>	<i>HEX</i>
3150	2	<i>Low penetration enable</i>	<i>0: disable, 1: enable (default 0)</i>	<i>UINT16</i>	<i>HEX</i>
3151	2	<i>High penetration enable</i>	<i>0: disable, 1: enable (default 0)</i>	<i>UINT16</i>	<i>HEX</i>
3152	2	<i>Reactive Mode</i>	<i>0: non-adjustable, 1: power factor, 2: reactive power (default 2)</i>	<i>UINT16</i>	<i>HEX</i>
3153	2	<i>Active anti-islanding</i>	<i>0: disable, 1: enable (default 0)</i>	<i>UINT16</i>	<i>HEX</i>

3154	2	Insulation detection	0: disable, 1: enable (default 0)	UINT16	HEX
3155	2	Priority	0: Active priority, 1: reactive priority (default 0)	UINT16	HEX
3156	2	PV power low shutdown	0: disable, 1: enable (default 1)	UINT16	HEX
3157	2	On/Off Command	0: On-grid, 1: Off-grid (default 0)	UINT16	HEX
3158	2	Grid-connected switching	0: Command Trigger, 1: Grid Trigger (default 0)	UINT16	HEX
3159	2	Grid-connected control mode	0: PQ, 1: MPPT, 2: CV, 3: VSG (default 0)	UINT16	HEX
3160	2	Off-grid control mode	0: VSG, 1: VF (default 1)	UINT16	HEX
3161	2	Reserved	0~65535 (default 0)	UINT16	HEX
3162	2	Reserved	0~65535 (default 0)	UINT16	HEX
3163	2	Reserved	0~65535 (default 0)	UINT16	HEX
3164	2	Reserved	0~65535 (default 0)	UINT16	HEX
3165	2	Authorized Capacity	100~3000 (10~300kW) (default 500)	UINT16	HEX
3166	2	Inertia time constant	0~10000 (0~100s) (default 500)	UINT16	HEX
3167	2	Damping factor	0~30000 (0~3000) (default 500)	UINT16	HEX
3168	2	Primary FM enable	0: disable, 1: enable (default 0)	UINT16	HEX
3169	2	Primary voltage regulation enable	0: disable, 1: enable (default 0)	UINT16	HEX
3170	2	Active FM coefficient	0~10000 (0~1000) (default 200)	UINT16	HEX
3171	2	FM frequency dead band	0~1000 (0~2Hz) (default 30)	UINT16	HEX
3172	2	FM active upper limit	0~1000 (0%~200%)	UINT16	HEX

			(default 1000)		
3173	2	FM active lower limit	-1000~0 (-200%~0%) (default -1000)	INT16	HEX
3174	2	Reactive regulator coefficient	0~1000 (0~100) (default 125)	UINT16	HEX
3175	2	Voltage deadband	0~1000 (0~100V) (default 50)	UINT16	HEX
3176	2	Upper limit of regulating reactive power	0~2000 (0%~200%) (default 1000)	UINT16	HEX
3177	2	Regulating reactive power lower limit	-2000~0 (-200%~0%) (default -1000)	INT16	HEX
3178	2	Charging method	0: two-part, 1: three-part (default 0)	UINT16	HEX
3179	2	Reserved	0~65535 (default 0)	UINT16	HEX
3180	2	Reserved	0~65535 (default 0)	UINT16	HEX
3181	2	Reserved	0~65535 (default 0)	UINT16	HEX
3182	2	Module address ⁽¹⁾	1~15 (default 1)	UINT16	HEX
3183	2	Cabinet address	1~10 (default 1)	UINT16	HEX
3184	2	Battery SOC	0~1000 (0%~100%) (default 0)	UINT16	HEX
3185	2	Battery SOC sag	0: Disabled, 1: Enabled (default 0)	UINT16	HEX
3186	2	Maximum apparent power limit	0~1200 (0%~120%) (default 1100)	UINT16	HEX
3187	2	Reactive power rate of change	1~30000 (1%/s~30000%/s) (default 2000)	UINT16	HEX
3188	2	Reactive power PT1 time constant	100~60000 (100ms~60000ms) (default 5000)	UINT16	HEX
3189	2	Reactive power change mode	0: Slope change, 1: PT1 change (Default: 0)	UINT16	HEX
3190	2	LVRT negative-sequence reactive current coefficient	0~600(0~6) (default 160)	UINT16	HEX

3191	2	<i>HVRT negative-sequence reactive current coefficient</i>	<i>0~600(0~6) (default 160)</i>	<i>UINT16</i>	<i>HEX</i>
3192	2	<i>VRT grid standard</i>	<i>0: China, 1: Europe, 2: Germany, 3: Austria, 4: Italy, 5: South Africa (default 0)</i>	<i>UINT16</i>	<i>HEX</i>
3193	2	<i>Over-frequency derating</i>	<i>0: Disabled, 1: Enabled (default 0)</i>	<i>UINT16</i>	<i>HEX</i>
3194	2	<i>Over-frequency derating slope</i>	<i>1~120 (0.1% ~ 12%) (default 50)</i>	<i>UINT16</i>	<i>HEX</i>
3195	2	<i>Over-frequency derating trigger frequency</i>	<i>1~500 (+0.01 ~ +5Hz) (default 20)</i>	<i>UINT16</i>	<i>HEX</i>
3196	2	<i>Over-frequency derating cut-off frequency</i>	<i>1~500 (+0.01 ~ +5Hz) (default 270)</i>	<i>UINT16</i>	<i>HEX</i>
3197	2	<i>Over-frequency derating trigger delay</i>	<i>0~20000 (ms) (default 0)</i>	<i>UINT16</i>	<i>HEX</i>
3198	2	<i>Over-frequency derating exit frequency enable</i>	<i>0: Disabled, 1: Enabled (default 0)</i>	<i>UINT16</i>	<i>HEX</i>
3199	2	<i>Over-frequency derating exit frequency</i>	<i>1~500 (+0.01 ~ +5Hz) (default 10)</i>	<i>UINT16</i>	<i>HEX</i>
3200	2	<i>Over-frequency derating exit delay</i>	<i>0~600 (s) (default 30)</i>	<i>UINT16</i>	<i>HEX</i>
3201	2	<i>Over-frequency derating power recovery gradient</i>	<i>6~3000 (%/min) (default 10)</i>	<i>UINT16</i>	<i>HEX</i>
3202	2	<i>Over-frequency derating power recovery mode</i>	<i>0: Fixed gradient, 1: Two-step gradient (default: 0)</i>	<i>UINT16</i>	<i>HEX</i>
3203	2	<i>Under-frequency</i>	<i>0: Disabled, 1: Enabled (default 0)</i>	<i>UINT16</i>	<i>HEX</i>

		<i>derating</i>			
3204	2	<i>Under-frequency derating slope</i>	<i>1~120 (0.1% ~ 12%) (default 50)</i>	<i>UINT16</i>	<i>HEX</i>
3205	2	<i>Under-frequency derating trigger frequency</i>	<i>-500~-1 (-5 ~ -0.01Hz) (default -20)</i>	<i>INT16</i>	<i>HEX</i>
3206	2	<i>Under-frequency derating cut-off frequency</i>	<i>-500~-1 (-5 ~ -0.01Hz) (default -270)</i>	<i>INT16</i>	<i>HEX</i>
3207	2	<i>Under-frequency derating trigger delay</i>	<i>0~20000 (ms) (default 0)</i>	<i>UINT16</i>	<i>HEX</i>
3208	2	<i>Under-frequency derating exit frequency enable</i>	<i>0: Disabled, 1: Enabled (default 0)</i>	<i>UINT16</i>	<i>HEX</i>
3209	2	<i>Under-frequency derating exit frequency</i>	<i>-500~-1 (-5 ~ -0.01Hz) (default -10)</i>	<i>INT16</i>	<i>HEX</i>
3210	2	<i>Undervoltage ramp-up exit delay</i>	<i>0~600 (s) (default 30)</i>	<i>UINT16</i>	<i>HEX</i>
3211	2	<i>Undervoltage ramp-up power recovery gradient</i>	<i>6~3000 (%/min) (default 10)</i>	<i>UINT16</i>	<i>HEX</i>
3212	2	<i>Undervoltage ramp-up power recovery mode</i>	<i>0: Fixed gradient, 1: Two-stage gradient (default: 0)</i>	<i>UINT16</i>	<i>HEX</i>
3213	2	<i>Frequency active power control reference power</i>	<i>0: Current power, 1: Rated power (default: 0)</i>	<i>UINT16</i>	<i>HEX</i>
3214	2	<i>P-U mode</i>	<i>0: Disabled, 1: Enabled (default 0)</i>	<i>UINT16</i>	<i>HEX</i>
3215	2	<i>P-U mode PT1 time constant</i>	<i>100~60000 (100ms~60000ms) (default 5000)</i>	<i>UINT16</i>	<i>HEX</i>
3216	2	<i>Discharge P-U Voltage Point 1</i>	<i>1000~1500 (100%~150%) (default 1100)</i>	<i>UINT16</i>	<i>HEX</i>
3217	2	<i>Discharge P-U</i>	<i>1000~1500 (100%~150%)</i>	<i>UINT16</i>	<i>HEX</i>

		<i>Voltage Point 2</i>	<i>(default 1110)</i>		
3218	2	<i>Discharge P-U Voltage Point 3</i>	<i>1000~1500 (100%~150%) (default 1120)</i>	<i>UINT16</i>	<i>HEX</i>
3219	2	<i>Discharge P-U Voltage Point 4</i>	<i>1000~1500 (100%~150%) (default 1130)</i>	<i>UINT16</i>	<i>HEX</i>
3220	2	<i>Discharge P-U Active Power Limit 1</i>	<i>0~1200 (0%~120%) (default 1000)</i>	<i>UINT16</i>	<i>HEX</i>
3221	2	<i>Discharge P-U Active Power Limit 2</i>	<i>0~1200 (0%~120%) (default 500)</i>	<i>UINT16</i>	<i>HEX</i>
3222	2	<i>Discharge P-U Active Power Limit 3</i>	<i>0~1200 (0%~120%) (default 0)</i>	<i>UINT16</i>	<i>HEX</i>
3223	2	<i>Discharge P-U Active Power Limit 4</i>	<i>0~1200 (0%~120%) (default 0)</i>	<i>UINT16</i>	<i>HEX</i>
3224	2	<i>Q-U Voltage Point 1</i>	<i>500~1500 (50%~150%) (default 920)</i>	<i>UINT16</i>	<i>HEX</i>
3225	2	<i>Q-U Voltage Point 2</i>	<i>500~1500 (50%~150%) (default 960)</i>	<i>UINT16</i>	<i>HEX</i>
3226	2	<i>Q-U Voltage Point 3</i>	<i>500~1500 (50%~150%) (default 1050)</i>	<i>UINT16</i>	<i>HEX</i>
3227	2	<i>Q-U active power point 4</i>	<i>500~1500 (50%~150%) (default 1080)</i>	<i>UINT16</i>	<i>HEX</i>
3228	2	<i>Q-U reactive power point 1</i>	<i>-1200~1200 (-120%~120%) (default 436)</i>	<i>INT16</i>	<i>HEX</i>
3229	2	<i>Q-U reactive power point 2</i>	<i>-1200~1200 (-120%~120%) (default 0)</i>	<i>INT16</i>	<i>HEX</i>
3230	2	<i>Q-U reactive power point 3</i>	<i>-1200~1200 (-120%~120%) (default 0)</i>	<i>INT16</i>	<i>HEX</i>
3231	2	<i>Q-U reactive power point 4</i>	<i>-1200~1200 (-120%~120%) (default -436)</i>	<i>INT16</i>	<i>HEX</i>
3232	2	<i>Q-U active power trigger value</i>	<i>0~1200 (0%~120%) (default 200)</i>	<i>UINT16</i>	<i>HEX</i>
3233	2	<i>Q-U active power exit value</i>	<i>0~1200 (0%~120%) (default 200)</i>	<i>UINT16</i>	<i>HEX</i>
3234	2	<i>Q-U power factor</i>	<i>0~1000 (0~1)</i>	<i>UINT16</i>	<i>HEX</i>

		<i>setpoint</i>	<i>(default 400)</i>		
3235	2	<i>PF-P active power point 1</i>	<i>0~1200 (0%~120%) (default 200)</i>	<i>UINT16</i>	<i>HEX</i>
3236	2	<i>PF-P Active Point 2</i>	<i>0~1200 (0%~120%) (default 500)</i>	<i>UINT16</i>	<i>HEX</i>
3237	2	<i>PF-P Active Point 3</i>	<i>0~1200 (0%~120%) (default 1000)</i>	<i>UINT16</i>	<i>HEX</i>
3238	2	<i>PF-P Active Point 4</i>	<i>0~1200 (0%~120%) (default 1000)</i>	<i>UINT16</i>	<i>HEX</i>
3239	2	<i>PF-P Power Factor Point 1</i>	<i>-1000~1000 (-1~1) (default 1000)</i>	<i>INT16</i>	<i>HEX</i>
3240	2	<i>PF-P Power Factor Point 2</i>	<i>-1000~1000 (-1~1) (default 1000)</i>	<i>INT16</i>	<i>HEX</i>
3241	2	<i>PF-P Power Factor Point 3</i>	<i>-1000~1000 (-1~1) (default -900)</i>	<i>INT16</i>	<i>HEX</i>
3242	2	<i>PF-P Power Factor Point 4</i>	<i>-1000~1000 (-1~1) (default -900)</i>	<i>INT16</i>	<i>HEX</i>
3243	2	<i>PF-P trigger voltage</i>	<i>50~150 (50%~150%) (default 110)</i>	<i>UINT16</i>	<i>HEX</i>
3244	2	<i>PF-P disengagement voltage</i>	<i>50~150 (50%~150%) (default 108)</i>	<i>UINT16</i>	<i>HEX</i>
3245	2	<i>Start-up reconnection time</i>	<i>0~300 (s) (default 60)</i>	<i>UINT16</i>	<i>HEX</i>
3246	2	<i>Fault reconnection time</i>	<i>0~600 (s) (default 60)</i>	<i>UINT16</i>	<i>HEX</i>
3247	2	<i>Fault reconnection power change rate</i>	<i>1~3000 (%/min) (default 100)</i>	<i>UINT16</i>	<i>HEX</i>
3248	2	<i>Start-up power change rate</i>	<i>1~3000 (%/min) (default 100)</i>	<i>UINT16</i>	<i>HEX</i>
3249	2	<i>Shutdown power change rate</i>	<i>1~30000 (%/s) (default 500)</i>	<i>UINT16</i>	<i>HEX</i>
3250	2	<i>Cumulative grid overvoltage</i>	<i>0: Disabled, 1: Enabled (default 0)</i>	<i>UINT16</i>	<i>HEX</i>
3251	2	<i>Cumulative grid</i>	<i>105~150 (105%~150%)</i>	<i>UINT16</i>	<i>HEX</i>

		<i>overvoltage point</i>	<i>(default 110)</i>		
3252	2	<i>Cumulative grid voltage time</i>	<i>1~20 (min) (default 10)</i>	<i>UINT16</i>	<i>HEX</i>
3253	2	<i>Cumulative grid overvoltage protection time</i>	<i>0~60000 (ms) (default 3000)</i>	<i>UINT16</i>	<i>HEX</i>
3254	2	<i>Q-U reactive offset</i>	<i>-1200~1200 (-120%~120%) (default 0)</i>	<i>INT16</i>	<i>HEX</i>
3255	2	<i>LVRT zero-current mode threshold</i>	<i>0~100 (0%~100%) (default 15)</i>	<i>UINT16</i>	<i>HEX</i>
3256	2	<i>Active power step change rate</i>	<i>1~30000 (1%/min~30000%/min) (default 20)</i>	<i>UINT16</i>	<i>HEX</i>
3257	2	<i>Active power change rate</i>	<i>0: seconds, 1: minutes (default 0)</i>	<i>UINT16</i>	<i>HEX</i>
3258	2	<i>QU curve deadband</i>	<i>0~10 (0%~10%) (default 0)</i>	<i>UINT16</i>	<i>HEX</i>
3259	2	<i>QU curve maximum voltage</i>	<i>50~150 (50%~150%) (default 104)</i>	<i>UINT16</i>	<i>HEX</i>
3260	2	<i>QU curve zero reactive voltage</i>	<i>50~150 (50%~150%) (default 100)</i>	<i>UINT16</i>	<i>HEX</i>
3261	2	<i>QU curve voltage offset</i>	<i>-20~20 (-20%~20%) (default 0)</i>	<i>INT16</i>	<i>HEX</i>
3262	2	<i>Minimum reactive power on the QU curve</i>	<i>-1200~0 (-120%~0%) (default -436)</i>	<i>INT16</i>	<i>HEX</i>
3263	2	<i>Active power point 1 on the Q-P curve</i>	<i>-1200~1200 (-120%~120%) (default 0)</i>	<i>INT16</i>	<i>HEX</i>
3264	2	<i>Active power point 2 on the Q-P curve</i>	<i>-1200~1200 (-120%~120%) (default 500)</i>	<i>INT16</i>	<i>HEX</i>
3265	2	<i>Active power point 3 on the Q-P curve</i>	<i>-1200~1200 (-120%~120%) (default 600)</i>	<i>INT16</i>	<i>HEX</i>
3266	2	<i>Active power point 4 on the Q-P curve</i>	<i>-1200~1200 (-120%~120%) (default 900)</i>	<i>INT16</i>	<i>HEX</i>
3267	2	<i>Active power point 5</i>	<i>-1200~1200</i>	<i>INT16</i>	<i>HEX</i>

		<i>on the Q-P curve</i>	<i>(-120%~120%) (default 1000)</i>		
3268	2	<i>Active power point 6 on the Q-P curve</i>	<i>-1200~1200 (-120%~120%) (default 1000)</i>	<i>INT16</i>	<i>HEX</i>
3269	2	<i>Reactive power point 1 on the Q-P curve</i>	<i>-1200~1200 (-120%~120%) (default 0)</i>	<i>INT16</i>	<i>HEX</i>
3270	2	<i>Reactive power point 2 on the Q-P curve</i>	<i>-1200~1200 (-120%~120%) (default 0)</i>	<i>INT16</i>	<i>HEX</i>
3271	2	<i>Reactive power point 3 on the Q-P curve</i>	<i>-1200~1200 (-120%~120%) (default -50)</i>	<i>INT16</i>	<i>HEX</i>
3272	2	<i>Q-P Reactive Point 4</i>	<i>-1200~1200 (-120%~120%) (default -436)</i>	<i>INT16</i>	<i>HEX</i>
3273	2	<i>Q-P Reactive Point 5</i>	<i>-1200~1200 (-120%~120%) (default -436)</i>	<i>INT16</i>	<i>HEX</i>
3274	2	<i>Q-P Reactive Point 6</i>	<i>-1200~1200 (-120%~120%) (default -436)</i>	<i>INT16</i>	<i>HEX</i>
3275	2	<i>Low Penetration Level 1 (Trigger Value)</i>	<i>1~100 (1%~100%) (default 90)</i>	<i>UINT16</i>	<i>HEX</i>
3276	2	<i>Low Penetration Level 2 (Curve Start Point)</i>	<i>1~100 (1%~100%) (default 90)</i>	<i>UINT16</i>	<i>HEX</i>
3277	2	<i>Low Penetration Level 3 (Curve End Point)</i>	<i>1~100 (1%~100%) (default 20)</i>	<i>UINT16</i>	<i>HEX</i>
3278	2	<i>Low-side short-circuit recovery</i>	<i>1~100 (1%~100%) (default 92)</i>	<i>UINT16</i>	<i>HEX</i>
3279	2	<i>High-side short-circuit Level 1 (trigger value)</i>	<i>100~150 (100%~150%) (default 110)</i>	<i>UINT16</i>	<i>HEX</i>
3280	2	<i>High-side short-circuit Level 2</i>	<i>100~150 (100%~150%) (default 120)</i>	<i>UINT16</i>	<i>HEX</i>
3281	2	<i>High-side short-circuit Level 3</i>	<i>100~150 (100%~150%) (default 130)</i>	<i>UINT16</i>	<i>HEX</i>
3282	2	<i>High-side</i>	<i>100~150 (100%~150%)</i>	<i>UINT16</i>	<i>HEX</i>

		<i>short-circuit recovery</i>	<i>(default 108)</i>		
3283	2	<i>Low-penetration Level 1 protection time</i>	<i>0~65000 (ms) (default 3000)</i>	<i>UINT16</i>	<i>HEX</i>
3284	2	<i>Low-penetration Level 2 protection time</i>	<i>0~65000 (ms) (default 3000)</i>	<i>UINT16</i>	<i>HEX</i>
3285	2	<i>Low-penetration Level 3 protection time</i>	<i>0~65000 (ms) (default 800)</i>	<i>UINT16</i>	<i>HEX</i>
3286	2	<i>High-penetration Level 1 protection time</i>	<i>0~65000 (ms) (default 15000)</i>	<i>UINT16</i>	<i>HEX</i>
3287	2	<i>High-penetration Level 2 protection time</i>	<i>0~65000 (ms) (default 2000)</i>	<i>UINT16</i>	<i>HEX</i>
3288	2	<i>High-penetration Level 3 protection time</i>	<i>0~65000 (ms) (default 500)</i>	<i>UINT16</i>	<i>HEX</i>
3289	2	<i>Charging P-U voltage point 1</i>	<i>500~1000 (50%~100%) (default 900)</i>	<i>UINT16</i>	<i>HEX</i>
3290	2	<i>Charging P-U voltage point 2</i>	<i>500~1000 (50%~100%) (default 890)</i>	<i>UINT16</i>	<i>HEX</i>
3291	2	<i>Charging P-U voltage point 3</i>	<i>500~1000 (50%~100%) (default 880)</i>	<i>UINT16</i>	<i>HEX</i>
3292	2	<i>Charging P-U voltage point 4</i>	<i>500~1000 (50%~100%) (default 870)</i>	<i>UINT16</i>	<i>HEX</i>
3293	2	<i>Charging P-U active power limit 1</i>	<i>-1200~0 (-120%~0%) (default -1000)</i>	<i>INT16</i>	<i>HEX</i>
3294	2	<i>Charging P-U active power limit 2</i>	<i>-1200~0 (-120%~0%) (default -500)</i>	<i>INT16</i>	<i>HEX</i>
3295	2	<i>Charging P-U active power limit 3</i>	<i>-1200~0 (-120%~0%) (default 0)</i>	<i>INT16</i>	<i>HEX</i>
3296	2	<i>Charging P-U active</i>	<i>-1200~0 (-120%~0%)</i>	<i>INT16</i>	<i>HEX</i>

		<i>power limit 4</i>	<i>(default 0)</i>		
3297	2	<i>Reserved</i>	<i>Reserved</i>	<i>UINT16</i>	<i>HEX</i>
3298	2	<i>Reserved</i>	<i>Reserved</i>	<i>UINT16</i>	<i>HEX</i>
3299	2	<i>Reserved</i>	<i>Reserved</i>	<i>UINT16</i>	<i>HEX</i>
3300	2	<i>Reserved</i>	<i>Reserved</i>	<i>UINT16</i>	<i>HEX</i>

Notes:

(1) The module address can only be adjusted in the non-parallel state using node number 160, and only IP65 modules support this feature.

4.5.2. Communication parameter setting

<i>ID</i>	<i>Byte</i>	<i>Item</i>	<i>Value</i>	<i>Value Type</i>	<i>Format</i>
3301	2	<i>Dry contact 1 default</i>	<i>"0: Normally open" "1:Normally closed"</i> <i>(default 0)</i>	<i>UINT16</i>	<i>HEX</i>
3302	2	<i>Dry contact 1 enable</i>	<i>"0: Disable" "1:Enable"</i> <i>(default 0)</i>	<i>UINT16</i>	<i>HEX</i>
3303	2	<i>Dry contact 2 default</i>	<i>"0: Normally open" "1:Normally closed"</i> <i>(default 0)</i>	<i>UINT16</i>	<i>HEX</i>
3304	2	<i>Dry contact 2 enable</i>	<i>"0: Disable" "1:Enable"</i> <i>(default 0)</i>	<i>UINT16</i>	<i>HEX</i>
3305	2	<i>BMS enable</i>	<i>"0: Disable" "1:Enable"</i> <i>(default 0)</i>	<i>UINT16</i>	<i>HEX</i>
3306	2	<i>BMS communication fault shutdown enable</i>	<i>"0: Disable" "1:Enable"</i> <i>(default 0)</i>	<i>UINT16</i>	<i>HEX</i>
3307	2	<i>IP Address 1</i>	<i>0~255 (default 192)</i>	<i>UINT16</i>	<i>HEX</i>
3308	2	<i>IP Address 2</i>	<i>0~255 (default 168)</i>	<i>UINT16</i>	<i>HEX</i>
3309	2	<i>IP Address 3</i>	<i>0~255 (default 1)</i>	<i>UINT16</i>	<i>HEX</i>
3310	2	<i>IP Address 4</i>	<i>0~255 (default 50)</i>	<i>UINT16</i>	<i>HEX</i>
3311	2	<i>Port number</i>	<i>0~65535 (default 502)</i>	<i>UINT16</i>	<i>HEX</i>
3312	2	<i>Subnet mask 1</i>	<i>0~255 (default 255)</i>	<i>UINT16</i>	<i>HEX</i>
3313	2	<i>Subnet mask 2</i>	<i>0~255 (default 255)</i>	<i>UINT16</i>	<i>HEX</i>

3314	2	Subnet mask 3	0~255 (default 255)	UINT16	HEX
3315	2	Subnet mask 4	0~255 (default 0)	UINT16	HEX
3316	2	Network Gateway 1	0~255 (default 192)	UINT16	HEX
3317	2	Network Gateway 2	0~255 (default 168)	UINT16	HEX
3318	2	Network Gateway 3	0~255 (default 1)	UINT16	HEX
3319	2	Network Gateway 4	0~255 (default 1)	UINT16	HEX
3320	2	MAC Address 1	0~255 (default 2)	UINT16	HEX
3321	2	MAC Address 2	0~255 (default 10)	UINT16	HEX
3322	2	MAC Address 3	0~255 (default 15)	UINT16	HEX
3323	2	MAC Address 4	0~255 (default 14)	UINT16	HEX
3324	2	MAC Address 5	0~255 (default 13)	UINT16	HEX
3325	2	MAC Address 6	0~255 (default 6)	UINT16	HEX
3326	2	Communication Failure Confirmation Time	5~300 (5~300s) (default 30)	UINT16	HEX
3327	2	Regulations	0: Australia A; 1: Australia B; 2: Australia C; 3: New Zealand; 4: Netherlands A; 5: Netherlands B; 6: Austria; 7: None; 8: China A; 9: China B; 10: Latvia A; 11: Latvia B (Default value: 7)	UINT16	HEX
3328	2	BMS Protocol	0: Gaote; 1: XieNeng; 2: Huasu; 3: Group Standard; 4: Ningde; 5: KeGong (Default value: 3)	UINT16	HEX
3329	2	Automatic Power-On	0: Disabled; 1: Enabled (Default value: 0)	UINT16	HEX
3330	2	Reserved	Reserved	UINT16	HEX
3331	2	Reserved	Reserved	UINT16	HEX
3332	2	Reserved	Reserved	UINT16	HEX
3333	2	Reserved	Reserved	UINT16	HEX
3334	2	Reserved	Reserved	UINT16	HEX

3335	2	Reserved	Reserved	UINT16	HEX

Notes:

(1) TCP related parameter changes must be powered down and rebooted to take effect.

4.5.3. Control parameter settings 2

ID	Byte	Item	Value	Value Type	Format
3350	2	Active Power Control Mode	0: Fixed slope, 1: Fixed frequency (default value is 0)	UINT16	HEX
3351	2	Frequency at which derating of zero discharge power occurs	1~500 (+0.01 ~ +5Hz) (default 75)	UINT16	HEX
3352	2	Frequency at which derating of maximum charging power occurs	1~500 (+0.01 ~ +5Hz) (default 200)	UINT16	HEX
3353	2	Frequency at which derating of zero charging power occurs	-500~-1 (-5 ~ -0.01Hz) (default -100)	INT16	HEX
3354	2	Frequency at which derating of maximum discharge power occurs	-500~-1 (-5 ~ -0.01Hz) (default -200)	INT16	HEX
3355	2	Active Power Control Frequency Observation Period	0~3600 (s) (default 600)	UINT16	HEX
3356	2	PF-P Reactive Power PT1 Time Constant	100~60000 (ms) (default 5000)	UINT16	HEX
3357	2	Grid Line Voltage AB Correction Coefficient	3686~4506 (default 4096)	UINT16	HEX
3358	2	Grid Line Voltage BC Correction Coefficient	3686~4506 (default 4096)	UINT16	HEX

3359	2	Grid Phase Voltage CN Correction Coefficient	3686~4506 (default 4096)	UINT16	HEX
3360	2	Reserved	Reserved	UINT16	HEX

4.5.4. Example of Telecommand

Example: Modbus node number 01, write telecommand message, ID 3050, value 100%, data order H-L, checksum order L-H.

Modbus RTU:

Telecommand (HEX): 01 06 0B EA 03 E8 AA A4

Answer (HEX): 01 06 0B EA 03 E8 AA A4

Modbus TCP:

Telecommand (HEX): 00 0C 00 00 00 06 01 06 0B EA 03 E8

Answer (HEX): 00 0C 00 00 00 06 01 06 0B EA 03 E8

4.6. Setting the date and time (remote control) - Function 03 or 16

(Reserved)

ID	Byte	Item	Value	Format
1500	2	hour	0 : 23	HEX
1501	2	minute	0 : 59	HEX
1502	2	seconds	0 : 59	HEX
1503	2	year	0000: 2099	HEX
1504	2	month	1 : 12	HEX
1505	2	day	1 : 31	HEX

4.7. Remote On/Off (Remote Control) - Function 05 or 06

ID	Byte	Item	Value	Value Type	Format
5050	2	Switching on and off	0xFF00: Power on; 0x0000: power off	UINT16	HEX
5051	2	Standby	0xFF00: Standby; 0x0000: Shutdown	UINT16	HEX

4.7.1. Remote control example

Example: Modbus node number 01, remote power-up command, single word write without parameters, data order H-L, checksum order L-H.

Modbus RTU:

Remote control (HEX): 01 05 13 BA FF 00 A9 5B

Answer (HEX): 01 05 13 BA FF 00 A9 5B

Modbus TCP:

Remote control (HEX): 00 10 00 00 00 06 01 05 13 BA FF 00

Answer (HEX): 00 10 00 00 00 06 01 05 13 BA FF 00